

UDON THANI, Thailand

WMO: 483540

Lat: 17.377N

Lon: 102.809E

Elev: 584

StdP: 14.39

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 56.1	(c) 59.2	(d) 43.9	(e) 43.3	(f) 69.7	(g) 47.9	(h) 50.4	(i) 69.2	(j) 12.2	(k) 75.8	(l) 11.2	(m) 76.8	(n) 1.2	(o) 90	(p) 0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 19.4	(c) 101.1	(d) 76.1	(e) 98.8	(f) 76.0	(g) 96.7	(h) 76.1	(i) 81.0	(j) 90.4	(k) 80.6	(l) 89.8	(m) 80.0	(n) 88.7	(o) 4.7	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	154.4	83.9	78.7	152.2	83.6	78.1	149.3	83.3	45.4	91.0	44.8	90.1	44.3	89.6	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.8	(b) 9.9	(c) 8.9	(e) 50.7	(f) 104.5	(g) 3.7	(h) 2.3	(i) 48.0	(j) 106.1	(k) 45.9	(l) 107.4	(m) 43.8	(n) 108.7	(o) 41.1	(p) 110.3		
(4) 11.8	(5) 9.9	(6) 8.9	(7) 50.7	(8) 104.5	(9) 3.7	(10) 2.3	(11) 48.0	(12) 106.1	(13) 45.9	(14) 107.4	(15) 43.8	(16) 108.7	(17) 41.1	(18) 110.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.3	74.3	78.2	83.1	86.5	85.4	84.7	83.6	83.0	82.7	81.6	78.4	74.0
	DBStd	5.47	4.61	5.31	5.02	4.17	3.29	2.48	2.54	2.20	2.06	2.69	3.87	4.88
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	7	3	1	0	0	0	0	0	0	0	0	0	3
	CDD50	11426	754	790	1026	1096	1098	1040	1042	1022	982	980	851	745
	CDD65	5960	292	371	562	646	633	590	577	557	532	515	402	283
	CDH74	65526	2562	3827	6739	8435	7613	6962	6481	5983	5598	5275	3679	2372
CDH80	28756	979	1844	3627	4707	3646	3012	2532	2149	1989	2033	1430	808	
Wind	WSAvg	3.0	2.8	2.9	3.2	3.2	3.1	3.2	3.4	3.3	2.8	2.8	2.8	2.9
Precipitation	PrecAvg	57.7	0.3	0.7	1.8	3.0	8.1	8.7	9.0	11.0	9.9	3.7	0.6	0.2
	PrecMax	95.2	2.8	4.5	8.2	8.6	16.5	25.8	21.1	22.6	25.3	11.6	5.0	2.3
	PrecMin	35.0	0.0	0.0	0.0	0.3	2.5	0.0	2.9	2.5	3.0	0.0	0.0	0.0
	PrecStd	11.8	0.6	0.8	1.9	1.9	3.5	3.7	3.9	4.4	4.6	2.8	1.0	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.5	98.6	102.8	104.9	101.9	97.2	96.6	94.1	93.6	93.5	94.4	92.1
		MCWB	71.5	71.4	73.6	76.8	78.8	79.3	79.4	79.5	79.1	78.1	75.4	74.9
	2%	DB	90.7	95.8	100.2	102.2	98.6	94.9	93.5	92.1	91.6	91.5	91.7	89.6
		MCWB	71.4	71.8	73.9	76.6	78.7	79.5	78.9	79.1	78.5	77.1	75.3	73.0
	5%	DB	88.2	93.4	97.7	99.7	95.8	93.0	91.5	90.3	90.1	90.0	89.6	87.4
		MCWB	70.7	72.4	74.0	76.8	78.6	79.1	78.6	78.7	78.1	76.3	74.3	71.6
	10%	DB	85.8	90.6	95.1	96.8	93.3	91.1	89.7	88.7	88.8	88.9	87.7	84.8
		MCWB	69.8	71.9	74.3	76.8	78.4	78.7	78.4	78.4	77.9	75.8	73.0	70.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.0	77.9	79.6	80.8	81.9	81.8	81.2	81.4	81.3	80.7	79.1	78.3
		MCDB	87.9	90.7	94.1	95.2	93.6	91.1	90.5	89.1	88.1	88.9	88.5	86.3
	2%	WB	74.2	76.1	78.0	79.8	80.8	80.9	80.3	80.5	80.4	79.4	77.6	75.1
		MCDB	85.4	88.9	92.1	94.1	92.1	89.8	88.6	88.1	87.2	87.0	86.0	84.8
	5%	WB	72.7	74.8	77.0	79.0	80.0	80.2	79.7	79.8	79.7	78.6	76.4	73.4
		MCDB	83.7	87.2	91.0	93.2	90.7	88.6	87.5	86.9	86.2	85.8	84.5	84.2
	10%	WB	71.1	73.5	76.0	78.2	79.4	79.7	79.1	79.2	79.1	77.9	75.4	72.0
		MCDB	82.2	85.4	89.3	91.6	89.3	87.5	86.4	85.7	85.2	84.6	83.2	82.2
Mean Daily Temperature Range	5% DB	MDBR	22.9	23.3	21.8	19.4	15.5	13.2	12.0	11.8	12.3	14.6	18.4	21.0
		MCDBR	25.1	26.1	25.7	23.3	19.3	15.9	14.9	14.2	14.5	16.0	19.6	21.9
	5% WB	MCWBR	8.8	7.8	7.0	5.8	5.2	4.8	4.7	4.7	4.6	4.5	5.9	7.5
		MCWBR	21.9	22.1	21.5	20.0	16.5	14.1	13.1	12.9	13.2	14.1	17.2	20.4
Clear-Sky Solar Irradiance	taub		0.474	0.582	0.677	0.636	0.509	0.460	0.447	0.463	0.508	0.544	0.466	0.467
	taud		2.084	1.818	1.645	1.778	2.152	2.314	2.347	2.287	2.145	2.003	2.198	2.122
	Ebn at Noon		256	233	216	227	254	265	268	266	255	240	257	253
	Edh at Noon		49	67	82	72	49	41	40	43	49	55	44	46
All-Sky Solar Radiation	RadAvg		1512	1630	1641	1784	1706	1583	1494	1461	1475	1563	1557	1466
	RadStd		98	84	86	78	108	107	93	94	90	101	53	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+4	N/A	N/A
(47) Regional (15 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	+2	N/A	N/A

Nomenclature: See separate page